

RAJKUMAR RAMANATHAN

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Machine Learning Engineer

Results-driven **Machine Learning Engineer** with a strong foundation in **deep learning, computer vision, and natural language processing (NLP)**. Skilled in **Python, TensorFlow, PyTorch, and scikit-learn**, with experience in **data preprocessing, feature engineering, and model optimization**. Proficient in building and deploying machine learning models for **object detection, resume parsing, and time series forecasting**. Hands-on experience with **YOLOv8, Faster R-CNN, Transformer models, and custom NLP pipelines using spaCy**. Strong knowledge of **data structures, algorithms, and cloud-based deployment (Azure)**. Passionate about solving real-world problems through AI and continuously learning advanced ML techniques.

Key Skills & Expertise:

- ✓ Machine Learning & Deep Learning (TensorFlow, PyTorch, scikit-learn)
- ✓ NLP (spaCy, Transformer-based models, Resume Parsing)
- ✓ Computer Vision (YOLOv8, Faster R-CNN, Object Detection)
- ✓ Time Series Forecasting & Predictive Analytics
- ✓ Data Preprocessing & Feature Engineering
- ✓ Model Deployment & Optimization (Docker, Flask, FastAPI, Microservices)
- ✓ Cloud Services (Azure, API Gateway)
- ✓ SQL & NoSQL Databases (MongoDB, PostgreSQL)
- ✓ Agile & Collaborative Development (Git, JIRA)

WORK EXPERIENCE

Relevantz Technology Services • Chennai • 05/2023 - Present

Junior Software Engineer • Full-time

- Developed and optimized machine learning models for computer vision, NLP, and time series forecasting, improving model efficiency and accuracy.
- Designed and implemented ID card detection using YOLOv8, integrating OCR for automated information extraction and storing results in MongoDB.
- Built a custom NLP model for resume parsing using spaCy, extracting key details such as name, experience, and skills with 85% accuracy.
- Worked on time series forecasting models for sales prediction, leveraging techniques such as Linear Regression, SDCA, and L-BFGS to improve forecast precision.
- Implemented **model deployment workflows** using **Docker**, enabling consistent environment management and smoother promotion from development to production. Integrated ML services with backend systems via **REST APIs**, supporting real-time and batch inference.
- Developed RESTful APIs and integrated ML models into web applications using **Flask, FastAPI, and Docker** for deployment.
- Collaborated with cross-functional teams using **Agile methodologies (Scrum, JIRA)** to deliver scalable and production-ready software solutions.
- Worked with **SQL and NoSQL databases (SQL, MongoDB)** for efficient data storage and retrieval in ML applications.
- Implemented **custom YOLOv8 modifications**, integrating **Ghost Bottleneck and CSPModule** to enhance object detection accuracy.
- Continuously explored advancements in **deep learning, cloud deployment (AWS/GCP), and transformer models** to improve model robustness and scalability.

EDUCATION

BE in COMPUTER SCIENCE

JP College of Engineering • Tenkasi • GPA: 8.78 • 06/2019 - 03/2023

CERTIFICATIONS

Hacker Rank Verified Software Engineer

Hacker Rank

Hacker Rank Verified Frontend Developer - (React)

Hacker Rank

MICROSOFT AI CLASSROOM SERIES CERTIFICATE

Microsoft

PROJECTS

TDS (ID Card Tailgate Detection System) (AI-Powered CCTV Detection)

Role : Machine Learning Engineer

Designed and delivered a **real-time security compliance platform** to detect **ID card violations and unauthorized tailgating events** using **computer vision and deep learning**. The system processes **live CCTV streams**, identifies whether individuals are wearing valid ID cards, detects multiple-person entry scenarios, and generates real-time alerts for security teams. The solution enables centralized monitoring, auditability, and analytics through an admin dashboard.

Key Responsibilities:

- Owned the end-to-end computer vision pipeline, including **raw dataset creation, image annotation, data cleaning, and preprocessing** for ID card and tailgating detection scenarios.
- Designed, trained, and optimized a **YOLOv8-based object detection model**, improving real-time ID compliance and unauthorized entry detection accuracy.
- Implemented **MLflow** for **model training, experiment tracking, hyperparameter comparison, and versioning**, enabling reproducible and production-ready ML workflows.
- Developed and deployed **FastAPI-based backend services**, exposing **REST APIs** for real-time inference, event logging, and dashboard integration.

AI Proctoring System (Computer Vision-Based Remote Exam Monitoring)

Relevantz Technology

Role : Machine Learning Engineer

Designed and developed an AI-powered online exam proctoring system to monitor candidates in real time and prevent malpractice during remote assessments. The system streams live webcam and audio data, performs real-time computer vision and audio analysis, and generates instant alerts for suspicious behavior. It detects violations such as multiple faces, head pose deviations, candidate absence, face occlusion, prohibited object usage, and abnormal audio activity. The platform includes a centralized dashboard for monitoring multiple candidates with on-demand camera and audio activation, real-time alerts, and secure transcription logging for audit and review purposes.

Key Responsibilities:

- Designed and implemented a **real-time webcam streaming pipeline** using **FastAPI, OpenCV, and MJPEG streaming**, optimized for multi-user monitoring.
- Built **AI-controlled camera lifecycle management**, enabling camera activation and deactivation via a **dashboard toggle**, preventing unnecessary camera usage and improving resource efficiency.
- Integrated **WebSocket-based real-time alerting system** to push AI violation events instantly from backend to frontend

iKYC – Intelligent AI-Powered Identity Verification Platform

Relevantz Technology

Role : Machine Learning Engineer

Developed a real-time AI-powered Deepfake Detection and Proctoring System to verify user authenticity during live camera sessions. The system performs continuous video streaming, deepfake classification, and behavioral analysis (eye

blink, face positioning, and head pose challenges) to prevent identity spoofing and impersonation attacks. The solution integrates a FastAPI backend with WebSocket-based real-time analysis delivery and a React-based dashboard for live monitoring and decision visualization.

Key Responsibilities:

- Designed and implemented a **real-time video streaming pipeline** using OpenCV and FastAPI, enabling seamless live camera feed processing and browser-based visualization.
- Built an **efficient custom deepfake detection model**, including data preprocessing, feature extraction, and optimized inference for real-time video frames.
- Integrated **ML model inference with confidence scoring**, generating REAL / FAKE classifications with threshold-based decision logic.
- Implemented **behavioral validation mechanisms** such as eye blink detection, face grid alignment, and randomized head pose challenges using MediaPipe.
- Developed a **WebSocket-based low-latency analysis channel** to stream AI predictions and behavioral validation results to the frontend dashboard in real time.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, FastAPI, React.js, Node.js

Databases: SQL Server, PostgreSQL, MongoDB, Cassandra

Cloud: Azure, AWS, GCP, IBM Cloud

Architecture: Microservices

DevOps/MLOps: Docker, Kubernetes, Nginx, Jenkins, Airflow

Computer Vision: OpenCV, YOLO, CNNs, TNNs, Efficient Net, Image Classification, Object Detection

Natural Language Processing: spaCy, NLTK, Transformers (BERT-based models)

Data Processing: NumPy, Pandas, PySpark